

Cross-Domain Fake Review Detection Using BERT with Linguistic and Behavioral Features

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Abstract

Fake reviews distort platform ratings and undermine consumer trust. This work investigates whether combining BERT text representations with linguistic and behavioral signals improves fake review detection across three distinct scenarios — a noisy Yelp dataset with platform-generated labels, the expert-labeled YelpChi dataset available only as pre-extracted behavioral features with no raw text, and a clean Amazon dataset of human-annotated reviews. For YelpChi, a Transformer Classifier trained exclusively on behavioral signals replaces BERT.

Results show that transformer-based models consistently outperform SVM baselines, but gains from text alone are modest on difficult datasets. Behavioral features deliver a 25.5% relative F1 gain on noisy Yelp, while BERT alone achieves near-perfect performance on clean Amazon ($F1=0.9765$, $AUC=0.9979$). For YelpChi, the Transformer Classifier achieves $F1=0.7004$ without any review text. Cross-domain experiments reveal a severe generalization gap — a model trained on Amazon collapses to $F1=0.5124$ on Yelp, demonstrating that models learn platform-specific patterns rather than universal deception signals. These findings establish that fake review detection is a multi-signal problem where behavioral patterns, label quality, and domain alignment matter as much as model architecture.

Keywords: Fake review detection, opinion spam, BERT, behavioral modeling, multi-signal detection, cross-domain evaluation, deception detection

1. Introduction

Online review platforms such as Yelp and Amazon play a major role in consumer decision making, with studies suggesting that more than 90% of purchasing decisions are influenced by online reviews [10]. This creates incentives for manipulation through fake reviews, with estimates suggesting 15–30% of online reviews may be deceptive.

Early detection approaches relied on lexical features and traditional classifiers such as Support Vector Machines [5], which struggle when deceptive reviews closely mimic genuine writing. Transformer models such as BERT [4] have improved text classification through contextual understanding, but deception is not always visible in language alone. Prior research shows that deceptive reviewers also exhibit abnormal behavioral patterns such as burst posting, extreme ratings, and single-review accounts, signals invisible to text-only models. Despite this, most prior work studies textual and behavioral signals independently and evaluates on a single platform, leaving their combined effect and cross-domain behavior largely unexplored.

Motivated by these gaps, this work investigates whether combining contextual text representations with linguistic and behavioral signals improves fake review detection across three datasets representing clean, noisy, and behavior-driven scenarios, including YelpChi where detection relies entirely on pre-extracted behavioral feature vectors. Cross-domain experiments examine whether models learn general deception patterns or platform-specific artifacts.

This study makes the following contributions. First, a multi-signal detection framework is evaluated across three datasets with model configurations adapted to the feature availability of each dataset. Second, a Transformer Classifier trained entirely on behavioral features is developed for YelpChi, demonstrating effective detection without any review text. Third, a systematic cross-domain evaluation reveals the generalization gap between platforms and the limitations of union training as a bridging strategy.

2. Related Work

Early research on fake review detection focused on linguistic characteristics of deceptive writing. Ott et al. [1] showed that fake reviews often contain exaggerated sentiment and fewer concrete experiential details. Traditional machine learning approaches such as Support Vector Machines [5] became standard baselines for text-based detec-

tion, while Jindal and Liu [11] demonstrated that opinion spam could be identified through textual and behavioral analysis.

Later research highlighted that textual signals alone are often insufficient when deceptive reviewers imitate genuine writing styles. Mukherjee et al. [3] showed that abnormal reviewer behavior can reveal coordinated spam activity. Similarly, Rayana and Akoglu [2] introduced the YelpChi dataset and demonstrated that behavioral network features significantly improve detection performance.

Advances in representation learning, from distributed word representations [6] to transformer architectures [7], culminated in BERT [4] which demonstrated strong performance through contextual embeddings. However, cross-domain robustness remains a challenge, as models trained on one platform often perform poorly on another [8]. Domain adaptation techniques such as domain adversarial training [9] attempt to address this limitation.

While prior work has explored textual and behavioral signals independently, fewer studies systematically compare their relative importance across datasets, examine cross-domain transfer, or evaluate detection in settings where no review text is available. This study addresses all three gaps.

3. Research Questions

Based on the limitations of text-only approaches, the lack of systematic cross-domain evaluation, and the unexplored setting of detection without review text, as identified in prior work, this study investigates the following research questions:

RQ1: How much improvement do transformer-based models provide over traditional text-based methods such as SVM for fake review detection?

RQ2: Do behavioral features provide additional benefits beyond textual representations, and can they serve as the sole detection signal when no review text is available, as in the YelpChi setting where only pre-extracted behavioral feature vectors are provided?

RQ3: How well do fake review detection models generalize across different platforms (cross-domain performance)?

RQ4: Does training on combined multi-platform data (union training) improve generalization, or does one domain dominate and hurt performance on the other?

RQ5: Does label noise in platform-generated datasets limit the effectiveness of transformer-based models?

4. Datasets

Three publicly available datasets were used, as summarized in Table 1, each representing a distinct detection scenario: Kaggle Yelp with noisy platform-generated labels and mild class imbalance, YelpChi with expert-labeled

behavioral features only and severe class imbalance, and Amazon with clean human-annotated. These datasets allow evaluation under varying levels of label noise, class imbalance, and feature availability.

4.1. Kaggle Yelp Dataset

The Kaggle Yelp dataset contains approximately 788,470 reviews [12] flagged by Yelp’s automated filtering algorithm, with 41.47% labeled as fake and 58.53% as genuine. As noted by Mukherjee et al. [3], platform-generated labels may contain noise, making this dataset representative of real-world detection conditions. Due to computational constraints, 50,000 samples per class were randomly selected to create a balanced subset. Three feature sets are available: raw review text, 15 handcrafted linguistic style features, and 13 behavioral features derived from reviewer and restaurant metadata including friend count, review count, useful and cool votes, fan count, review rating, restaurant rating, and filtered review count.

4.2. YelpChi Dataset

The YelpChi dataset introduced by Rayana and Akoglu [2] contains 45,954 reviews with approximately 14.5% fake and 85.5% genuine reviews. Unlike the other two datasets, it provides only pre-extracted behavioral feature vectors in a .mat format without any raw review text, constructed by the original authors using reviewer activity and network-based signals [2]. This makes it the most constrained detection setting, requiring a Transformer Classifier trained exclusively on behavioral signals. The dataset exhibits strong class imbalance at approximately 6:1, requiring imbalance-aware training strategies.

4.3. Amazon Dataset

The Amazon dataset contains approximately 40,000 product reviews [14] with a balanced 50/50 distribution across multiple product categories. Each review includes text, a binary label, rating, and product category. Three feature sets are available: raw review text, 15 handcrafted linguistic style features, and 8 behavioral features derived from review-level signals including star rating, category encoding, review length, and linguistic activity indicators. Because fake reviews are computer-generated, they often contain detectable linguistic artifacts, making this dataset the most suitable for text-based detection and serving as an approximate upper bound for text-only model performance.

5. Technical Approach

5.1. SVM Baseline

TF-IDF converts raw review text into feature vectors for a linear kernel SVM classifier [5]. For YelpChi, the SVM baseline is trained on pre-extracted behavioral features with

standard scaling and class-balanced weighting, consistent with the Transformer Classifier.

5.2. BERT Text Model

BERT-base-uncased [4] is fine-tuned for binary classification. The CLS embedding is passed to a classification head with dropout.

5.3. BERT with Linguistic Features

Extends BERT [4] by incorporating 15 linguistic style features including word count, lexical diversity, punctuation density, and POS tag ratios. Features are projected through a dedicated branch with batch normalization, ReLU, and dropout, then concatenated with the CLS embedding before classification. Evaluated on Kaggle Yelp and Amazon only, as YelpChi contains no raw text.

5.4. BERT with Behavioral Features

Extends BERT [4] with reviewer activity features. For Kaggle Yelp, 13 reviewer and business metadata features are used. For Amazon, 8 review-level behavioral signals are used. Features are projected through the same branch design and concatenated with the CLS embedding.

5.5. Transformer Classifier for YelpChi

Since YelpChi [2] provides no raw review text, a transformer-style classifier is trained on behavioral feature vectors. The architecture consists of an input projection layer, four residual feed-forward blocks with layer normalization and GELU activation, and a two-layer classification head. Training configuration details are provided in Section 6.

5.6. Implementation Details

All models were implemented using PyTorch 2.10.0 with CUDA 12.8 on Google Colab using an NVIDIA A100 GPU. Pre-trained BERT weights were loaded via HuggingFace Transformers [13]. Linguistic features were extracted using NLTK [15]. All experiments used a fixed random seed. Full details are provided in Appendix C.

6. Experiment Setup

This section describes the experimental design, training configuration, and evaluation protocol used across all model configurations and datasets.

6.1. Data Preprocessing

Reviews with missing values were removed. Text was tokenized using the BERT tokenizer [4] and truncated to 128 tokens. For SVM [5], TF-IDF unigram and bigram features were used. YelpChi [2] behavioral feature vectors were used directly from the provided .mat file. For Kaggle

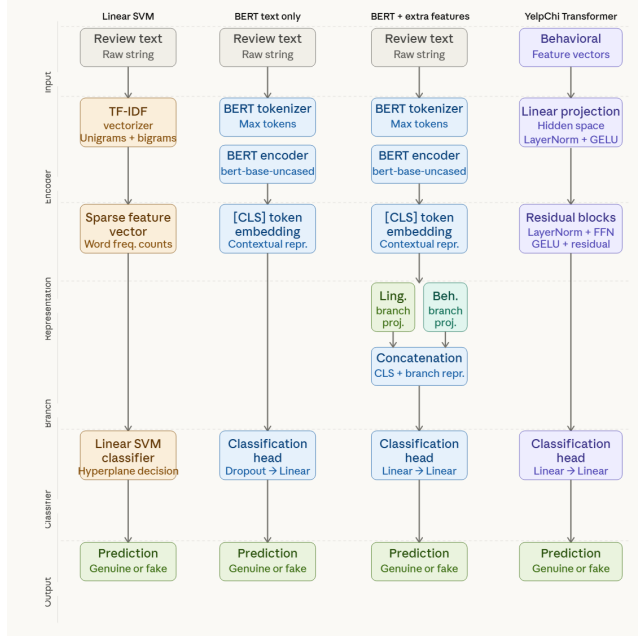


Figure 1: Overview of the four model configurations: Linear SVM baseline (left), BERT text-only (center left), BERT with additional feature branches — linguistic or behavioral — tested separately (center right), and YelpChi Transformer Classifier trained entirely on behavioral features (right).

Yelp [12], 50,000 samples per class were randomly selected to create a balanced subset.

Linguistic features were extracted from raw text before BERT cleaning to preserve surface signals. Behavioral features were derived from reviewer metadata. All non-text features were standardized using StandardScaler, with RobustScaler applied for Amazon [14] linguistic features to reduce sensitivity to outliers. All preprocessing was fitted on training data only to prevent leakage.

6.2. Data Splitting

Stratified 80/10/10 train/validation/test splits were applied across all datasets to preserve class distributions. For the YelpChi SVM baseline, an 80/20 split was used without a validation set. All preprocessing transformations were fitted only on training data.

6.3. Training Configuration

BERT-based models were trained using AdamW with batch size 32 and weight decay 0.01, with learning rates of 2×10^{-5} for BERT parameters and 1×10^{-3} for task-specific branches. A linear warmup over 10% of steps was followed by linear decay with gradient clipping at norm 1.0. Amazon models trained for up to 3 epochs and Kaggle Yelp

Dataset	Size	Fake%	Label Type	Text	Linguistic	Behavioral	Purpose
Kaggle Yelp	~788K	41.47%	Algorithm-generated	Yes	15 Features	13 Features	Real-world evaluation
YelpChi	45,954	14.5%	Expert-labeled	No	No	Pre-extracted	Behavioral analysis
Amazon	~40K	50%	Human-annotated	Yes	15 Features	8 Features	Clean benchmark

Table 1: Summary of datasets used in this study, showing size, class distribution, label quality, and evaluation purpose.

for 5, both with early stopping at patience 2 based on validation F1.

The YelpChi Transformer Classifier used AdamW with batch size 256, learning rate 1×10^{-3} , and weight decay 1×10^{-4} , trained for up to 20 epochs with early stopping at patience 3.

Weighted cross-entropy with square-root class weighting was applied for YelpChi and Kaggle Yelp:

$$w_i \propto \text{normalize}\left(\frac{1}{\sqrt{f_i}}\right) \quad (1)$$

where f_i is the class proportion in training data. Standard unweighted loss was used for Amazon.

6.4. Evaluation Metrics

Performance was evaluated using macro F1, ROC-AUC, and accuracy. Macro F1 is the primary metric as it treats both classes equally regardless of frequency, making it appropriate for imbalanced datasets. Confusion matrices were analyzed to examine false positive and false negative trade-offs.

6.5. Cross-Domain Setup

Cross-domain experiments used the BERT text-only model [4] to isolate domain shift from feature fusion. Models trained on Amazon [14] were evaluated on Yelp [12] and vice versa. A union training experiment tested whether combined data improves generalization. YelpChi [2] was excluded as it contains no raw review text, making it incompatible with the BERT text-only model.

6.6. Hyperparameter Tuning and Threshold Selection

Hyperparameters were selected based on validation F1. For YelpChi models, the classification threshold was tuned by searching 0.05 to 0.60 in increments of 0.01, selecting the value maximizing macro F1. The optimal threshold was consistently below 0.5, reflecting the model’s tendency to underestimate the minority fake class under severe imbalance. Default threshold of 0.5 was retained for Kaggle Yelp and Amazon as both use balanced training subsets.

7. Experiment Results and Analysis

This section presents the experimental results across all model configurations and datasets, summarized in Table 2, followed by cross-domain and union training experiments.

7.1. Comparative Model Performance Across Datasets

Table 2 summarizes performance across all configurations.

Kaggle Yelp. BERT improves only marginally over SVM (0.620 to 0.6429) due to label noise. Linguistic features add negligible gain (+0.006), likely because BERT already encodes the surface-level style signals they capture, and behavioral features produce the largest gain in the study, raising F1 to 0.8077, a 25.5% relative gain confirming that reviewer activity signals detect what text cannot.

YelpChi. Despite 79.6% accuracy, SVM detects only 11.2% of actual fake reviews, confirming accuracy is misleading under severe class imbalance. The Transformer Classifier substantially recovers fake recall to 72.0% (F1=0.7004), with threshold tuning further improving to F1=0.7591 and AUC=0.8821.

Amazon. Computer-generated fakes contain detectable linguistic artifacts, giving SVM a strong baseline (F1=0.920). BERT reaches near-perfect performance (F1=0.9765, AUC=0.9979), and additional features produce no gain, confirming redundancy rather than complementarity at this performance ceiling.

Across all three datasets, transformer-based models outperform SVM in every configuration, but improvement magnitude depends on data characteristics. Text alone suffices on clean Amazon, behavioral signals prove decisive on noisy Yelp, and the Transformer Classifier outperforms SVM on YelpChi without any text. Linguistic features provide no meaningful gain in any configuration. Since behavioral features are manually engineered for Kaggle Yelp but expert-constructed for YelpChi, gains across these datasets should be interpreted independently.

7.2. Evaluation Metrics of Best Performing Models

Evaluation metrics of Best performing models is summarized in Table 3. BERT + Behavioral on Kaggle Yelp achieves balanced precision and recall of 0.81 (F1=0.8077,

Dataset	Model	F1	AUC	Acc.	Observation
Kaggle Yelp	SVM (TF-IDF)	0.62	0.703	0.650	Baseline performance
Kaggle Yelp	BERT	0.643	0.701	0.645	Context improves slightly
Kaggle Yelp	BERT + Linguistic	0.649	0.704	0.650	Small improvement
Kaggle Yelp	BERT + Behavioral	0.808	0.887	0.808	Largest gain
YelpChi	SVM	0.647	0.774	0.796	Weak minority detection
YelpChi	Transformer Classifier	0.700	0.859	0.808	Better recall
YelpChi	Transformer Classifier + Weighted Loss	0.759	0.882	0.878	Best performance
Amazon	SVM	0.920	0.975	0.920	Strong text signal
Amazon	BERT	0.977	0.998	0.977	Near perfect
Amazon	BERT + Behavioral	0.976	0.999	0.976	No added benefit
Amazon	BERT + Linguistic	0.970	0.998	0.970	Slight degradation — redundant with BERT

Table 2: Performance of all models across datasets. Macro F1 is the primary metric. YelpChi results include weighted loss and threshold tuning.

Dataset	Model	Precision	Recall	F1 Macro	Accuracy	ROC-AUC
Kaggle Yelp	BERT + Behavioral	0.81	0.81	0.8077	0.8077	0.8872
YelpChi	Transformer Classifier + Weighted Loss	0.76	0.76	0.7591	0.8777	0.8821
Amazon	BERT	0.98	0.98	0.9765	0.9765	0.9979

Table 3: Precision, recall, macro F1, accuracy, and ROC-AUC of the best performing model on each dataset.

AUC=0.8872), confirming behavioral features improve detection of both classes equally. The YelpChi Transformer Classifier achieves AUC=0.8821 and accuracy=0.8777, but macro F1=0.7591 reflects the difficulty of minority class detection under severe imbalance. On Amazon, BERT alone achieves precision and recall of 0.98 (F1=0.9765, AUC=0.9979), confirming text representations are sufficient when fake reviews are linguistically distinct. The variation in macro F1 across datasets reflects differences in data quality and class balance rather than model limitations.

7.3. Confusion Matrix Analysis

Dataset	Model	TN	FP	FN	TP
Kaggle Yelp	BERT + Behavioral	4007	993	930	4070
YelpChi	Transformer (tuned)	3661	287	279	389
Amazon	BERT	1940	82	13	2009

Table 4: Confusion matrix values for the best performing model on each dataset. TN = True Negatives, FP = False Positives, FN = False Negatives, TP = True Positives.

Confusion matrices reveal dataset-dependent behavior. BERT + Behavioral achieves balanced detection on Kaggle Yelp (TP=4070, TN=4007, FN=930). The YelpChi Transformer Classifier correctly identifies 389 out of 668 fake re-

views despite severe 6:1 class imbalance. BERT achieves near-perfect classification on Amazon with only 13 false negatives out of 2,022 fake reviews. The declining false negative count across datasets — 930, 279, and 13 — directly reflects increasing label quality and data cleanliness.

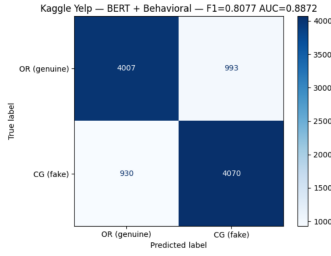
7.4. Cross-Domain Results

Table 5 summarizes cross-domain and union training results (macro F1, ROC-AUC, and accuracy). All experiments use the BERT text-only model to isolate domain shift effects.

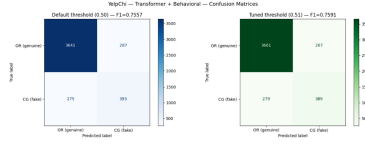
Table 5: Cross-domain and union training results.

Training Data	Test Data	F1	AUC	Acc.
Amazon	Yelp	0.512	0.519	0.516
Yelp	Amazon	0.753	0.832	0.748
Union	Yelp	0.621	0.666	0.623
Union	Amazon	0.973	0.996	0.971

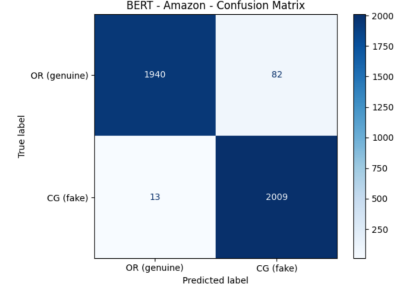
Cross-domain results reveal strong platform dependence. A model trained on Amazon performs near-randomly on Yelp (F1=0.5124, AUC=0.5188), confirming that patterns learned from computer-generated synthetic fakes do not transfer to human-written deceptive content. The reverse direction generalizes substantially better (Yelp to Ama-



(a) Kaggle Yelp — BERT + Behavioral
(F1 = 0.808, AUC = 0.887)



(b) YelpChi — Transformer + Behavioral
(F1 = 0.759, threshold = 0.51)



(c) Amazon — BERT text-only
(F1 = 0.977, AUC = 0.998)

Figure 2: Confusion matrices of the best-performing models: BERT + Behavioral (Kaggle Yelp), Transformer classifier (YelpChi), and BERT (Amazon).

zon F1=0.7531), because training on diverse human-written noisy data produces broader representations than training on narrow synthetic artifacts. Union training preserves Amazon performance (F1=0.9725) but degrades Yelp performance (F1=0.6207 vs Yelp-only F1=0.6429), as Amazon’s cleaner labels and balanced classes produce stronger gradients that dominate joint optimization. Simply combining datasets is insufficient to bridge the domain gap.

7.5. Research Question Answers

RQ1: Transformer models outperform SVM across all datasets: Kaggle Yelp (0.620 to 0.6429), Amazon (0.920 to 0.9765), and YelpChi (0.647 to 0.7004). Gains are marginal on noisy Kaggle Yelp, confirming limits of contextual text under label noise.

RQ2: Behavioral features provide the largest gains on Kaggle Yelp (0.6429 to 0.8077, +25.5% relative gain). YelpChi improves from F1=0.7004 to 0.7591 with threshold tuning. No benefit is observed on Amazon.

RQ3: Cross-domain results show strong asymmetry. Amazon to Yelp performs near randomly (F1=0.5124), while Yelp to Amazon achieves F1=0.7531, indicating better transfer from harder real-world data.

RQ4: Union training preserves Amazon performance (F1=0.9725) but degrades Yelp performance (F1=0.6207 vs F1=0.6429), indicating the cleaner Amazon signal dominates joint training.

RQ5: BERT achieves F1=0.6429 on noisy Kaggle Yelp versus F1=0.9765 on clean Amazon. YelpChi achieves F1=0.7004 without any review text, confirming that label quality and supervision signal matter as much as model architecture.

8. Discussion and Significance

A central lesson from this work is that model architecture alone does not determine performance — the nature of the training signal matters equally. This challenges the assumption that stronger pre-trained representations consistently improve downstream performance regardless of data quality.

On Kaggle Yelp, BERT barely improved over SVM (F1: 0.620 to 0.6429) despite its bidirectional attention mechanism, confirming that unreliable supervision pushes gradient updates toward noise memorization. The 25.5% F1 gain from behavioral features confirms the discriminating signal resided in reviewer activity patterns rather than text. YelpChi reinforces this — a Transformer Classifier achieves F1=0.7004 and 72% fake recall without any text input, demonstrating that behavioral signals carry independent discriminative power reflecting the quality of expert-constructed features.

Cross-domain experiments reveal that fine-tuning causes BERT to overfit to platform-specific lexical patterns, as evidenced by the Amazon→Yelp collapse to F1=0.5124, while the reverse achieves F1=0.7531, suggesting that training on harder real-world data acts as an implicit regularizer. Union training degrades Yelp performance from 0.6429 to 0.6207 while preserving Amazon performance, consistent with the cleaner Amazon distribution dominating joint optimization, motivating domain adversarial training as a more principled alternative.

Finally, standard accuracy is deeply misleading on imbalanced datasets. On YelpChi, a majority-class predictor achieves 85.5% accuracy while detecting zero fake reviews, confirming that macro F1, threshold tuning, and stratified evaluation are essential for rigorous evaluation.

9. Limitations

This study has several limitations. First, the datasets differ in size, labeling quality, and feature availability, making direct comparison challenging. Behavioral features also differ in construction — manually engineered for Kaggle Yelp and pre-extracted by the original authors [2] for YelpChi using network-based methodology — and therefore behavioral gains across these two datasets should be interpreted independently rather than as a direct comparison.

Second, behavioral features for Kaggle Yelp and Amazon were limited to manually engineered signals, leaving more complex relationships such as reviewer networks and temporal posting patterns unexplored. Third, cross-domain experiments were limited to two platforms, and finally, computational constraints limited hyperparameter exploration and large-scale experimentation.

10. Conclusion

This study investigated whether combining transformer-based text representations with behavioral signals improves fake review detection across platforms. Transformer-based models consistently outperform SVM baselines across all three datasets. For YelpChi, a Transformer Classifier trained on behavioral features achieves $F1=0.7004$ and 72% fake recall without any text input. Behavioral signals deliver a 25.5% relative $F1$ gain on noisy Kaggle Yelp, while BERT alone achieves near-perfect performance on clean Amazon ($F1=0.9765$), confirming that feature augmentation is conditional rather than universal.

Cross-domain experiments reveal that models learn platform-specific artifacts rather than transferable deception signals. Near-random Amazon to Yelp transfer ($F1=0.5124$) and the failure of union training confirm that domain alignment matters as much as model architecture.

Fake review detection is fundamentally a multi-signal problem. Label quality, behavioral patterns, and domain alignment matter as much as model choice. Behavioral signals should be treated as first-class features, and domain-specific training is a fundamental requirement for reliable cross-platform deployment.

11. Future Work

Several directions extend this work:

- A unified triple-fusion model combining textual, linguistic, and behavioral features could clarify whether behavioral signals remain dominant when all three are available simultaneously.
- Domain adversarial training could address the cross-domain failure mode by forcing the encoder to learn domain-invariant representations rather than platform-specific artifacts.

- Graph Neural Networks and temporal convolutional networks could capture reviewer-business relationships and burst posting patterns that static behavioral features miss.
- Large language models could be evaluated as zero-shot detectors, offering interpretability through natural language explanations that black-box classifiers cannot provide.

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Appendix

Linguistic Features (15) — Kaggle Yelp and Amazon

Extracted from raw review text before BERT tokenization. Feature names match exactly as used in the notebooks.

No.	Feature Name	Description
1	review_len	Total character length of the review
2	word_count	Total number of words in the review
3	sentence_count	Total number of sentences in the review
4	avg_word_len	Mean number of characters per word
5	avg_sentence_len	Mean number of words per sentence
6	exclamation_count	Total number of exclamation marks
7	question_count	Total number of question marks
8	capital_ratio	Proportion of uppercase characters to total characters
9	stopword_ratio	Proportion of stop words to total words
10	unique_ratio	Ratio of unique words to total words
11	first_person_ratio	Proportion of first-person pronouns to total words
12	third_person_ratio	Proportion of third-person pronouns to total words
13	noun_ratio	Proportion of nouns to total words via POS tagging
14	adj_ratio	Proportion of adjectives to total words via POS tagging
15	adv_ratio	Proportion of adverbs to total words via POS tagging

(a) Linguistic features

Behavioral Features — Kaggle Yelp (13)

Derived from reviewer and restaurant metadata CSVs. Feature names match exactly as used in the *Yelp_BERT_Behavioral* notebook.

No.	Feature Name	Description
1	friendCount	Number of friends the reviewer has
2	reviewer_review_count	Total number of reviews posted by the reviewer
3	firstCount	Number of first reviews posted by the reviewer
4	usefulCount	Number of useful votes received by the reviewer
5	coolCount	Number of cool votes received by the reviewer
6	funnyCount	Number of funny votes received by the reviewer
7	complimentCount	Number of compliments received by the reviewer
8	tipCount	Number of tips posted by the reviewer
9	fanCount	Number of fans the reviewer has
10	review_rating	Star rating given in the review
11	rest_rating	Overall rating of the restaurant
12	rest_review_count	Total number of reviews for the restaurant
13	filReviewCount	Number of filtered reviews for the restaurant

(b) Behavioral features

Appendix — Implementation Details

Parameter	Value
Framework	PyTorch 2.0.1
CUDA Version	12.5
GPU	NVIDIA A100 80GB 40GB
GPU Memory	70.1 GB
Package	TensorFlow
BERT Model	bert-base-uncased
Transformer Library	HuggingFace Transformers
Feature Extraction	ReLU
Optimizer	AdamW
BERT Learning Rate	2e-5
Dropout Learning Rate	1e-5
Batch Size	16
Max Epochs	4
Early Stopping Patience	2
Max Sequence Length	512 tokens
Warmup Steps	10% of total steps
Gradient	0.1
Gradient Clipping	1.0
Cross Entropy	Ignore loss method
Transformer Seed	803014
Random Seed	Fixed for reproducibility
Threshold Tuning	Validation set range 0.40-0.60

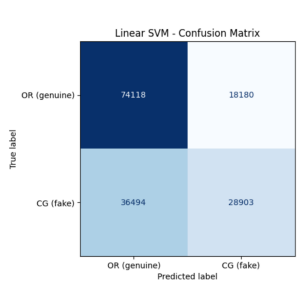
(c) Implementation details

Appendix C — Architecture Details

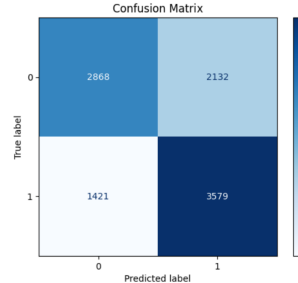
Exact layer configurations for all model components. Dimensions in parentheses indicate input→output sizes.

Parameter	Value
BERT encoder	bert-base-uncased, 12 layers, 768 hidden dims
Linguistic branch	Linear(15→128) → BatchNorm → ReLU → Dropout → Linear(128→64) → ReLU
Behavioral branch (Kaggle Yelp)	Linear(13→64) → BatchNorm → ReLU → Dropout → Linear(64→64) → ReLU
Behavioral branch (Amazon)	Linear(8→64) → BatchNorm → ReLU → Dropout → Linear(64→64) → ReLU
Fusion classifier	Linear(832→256) → ReLU → Dropout → Linear(256→2)
YelpChi input projection	Linear(input→256) → LayerNorm → GELU → Dropout
YelpChi residual blocks	4 blocks: LayerNorm → Linear(256→1024) → GELU → Dropout → Linear(1024→256) → Dropout
YelpChi classifier	LayerNorm → Dropout → Linear(256→64) → GELU → Dropout → Linear(64→2)

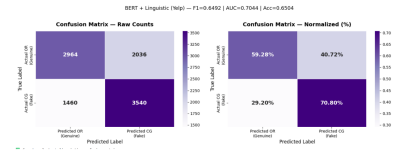
(d) Architecture details



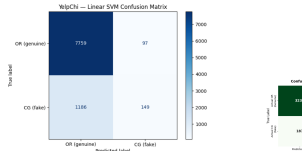
(e) Kaggle Yelp SVM (F1 = 0.620)



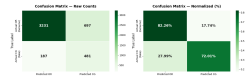
(f) Kaggle Yelp BERT (F1 = 0.643)



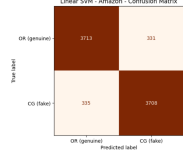
(g) Kaggle Yelp BERT+Ling (F1 = 0.649)



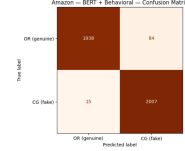
(h) YelpChi SVM (F1 = 0.647)



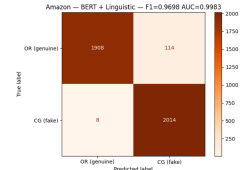
(i) YelpChi Transformer (F1 = 0.700)



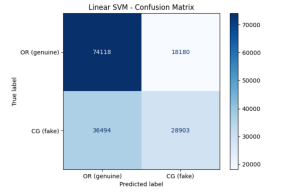
(j) Amazon SVM (F1 = 0.920)



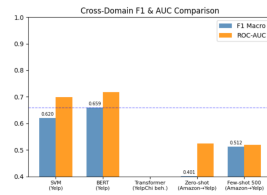
(k) Amazon BERT+Beh (F1 = 0.976)



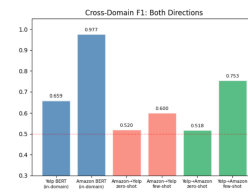
(l) Amazon BERT+Ling (F1 = 0.970)



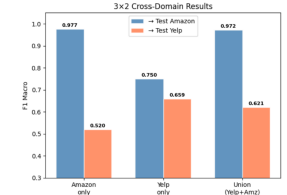
(m) SVM on Yelp



(n) Amazon→Yelp (F1 = 0.512)



(o) Yelp→Amazon (F1 = 0.753)



(p) Union training

Figure 3: Appendix: Row 1 — feature lists, implementation details, and architecture. Row 2 — Kaggle Yelp confusion matrices (SVM, BERT, BERT+Linguistic). Row 3 — YelpChi (left two) and Amazon (right three) confusion matrices. Row 4 — cross-domain and union training results.